



KTH Computer Science
and Communication

Exam in DD2421 Machine Learning

2025-03-12, kl 8.00 – 12.00

Aids allowed: *calculator, language dictionary.*

To take this exam you must be registered to this specific exam as well as to the course.

In order to pass this exam, your score x first needs to be 20 or more (out of 42, full point). In addition, given your points y from the Programming Challenge (out of 18, full point), the requirements on the total points, $p = x + y$, are preliminarily set for different grades as:

$$54 < p \leq 60 \rightarrow A$$

$$48 < p \leq 54 \rightarrow B$$

$$42 < p \leq 48 \rightarrow C$$

$$36 < p \leq 42 \rightarrow D$$

$$29 < p \leq 36 \rightarrow E \text{ (A pass is guaranteed with the required points for 'E'.)}$$

$$0 \leq p \leq 29 \rightarrow F$$

This exam consists of sections **A**, **B**, and **C**.

NB: use different papers (answer sheets) for different sections A, B, and C.

This means you shall start a new page after section A, and again after section B.

A Graded problems

Potential inquiries to be addressed to Atsuto Maki.

A-1 Terminology

(5p)

For each term (a–e) in the left list, find the explanation from the right list which *best* describes how the term is used in machine learning.

- | | |
|-------------------------------|--|
| | 1) Training samples representing a class |
| | 2) An approach to fit a model to data with outliers |
| | 3) A technique for margin maximization |
| a) Occams's razor | 4) Algorithm to learn with latent variables |
| b) k -fold cross validation | 5) A technique used in train artificial neural networks |
| c) OOD examples | 6) Robust method for Monte Carlo tree search |
| d) RANSAC | 7) A technique for assessing a model by exploiting available data for training and testing |
| e) Dropout | 8) A principle to choose the simplest explanation |
| | 9) An approach to find useful dimension for classification |
| | 10) To be rejected as "don't know" category |

A-2 Nearest Neighbor, Classification

(4p)

Suppose that we take a data set, divide it into training and test sets, and then try out two different classification procedures. We use *three-quarters* of the data for training, and the remaining *one-quarter* for testing. First we use *Bagging* and get an error rate of 10% on the training data. We also get the average error rate (weighted average over both test and training data sets) of 12%. Next we use 1-nearest neighbor and get an average error rate (weighted average over both test and training data sets) of 6%.

- a) What was the error rate with 1-nearest neighbor on the training set? (1p)
- b) What was the error rate with 1-nearest neighbor on the test set? (1p)
- c) What was the error rate with Bagging on the test set? (1p)
- d) Based on these results, indicate the method which we should prefer to use for classification of new observations, with a simple reasoning. (1p)

A-3 Entropy and Decision Trees/Forests

(5p)

- a) Consider a single toss of *skewed* coin (it is likely to show one side more than the other side). Regarding the uncertainty of the outcome {head, tail}, which one of the following is correct?

- i. the entropy is smaller than one bit.
- ii. the entropy is equal to one bit.
- iii. the entropy is greater than two bits.

Motivate your answer by short phrases. (1p)

- b) Indicate any valid ones as the basic strategy for selecting a question (attribute) at each node in decision trees.

- i. To maximize the expected reduction of the entropy.
- ii. To minimize the expected reduction of the entropy.
- iii. To minimize the expected reduction of gini impurity.

Motivate your answer by short phrases. (1p)

- c) How is Shannon's information content defined for an event i with probability $P(i)$? (1p)

- d) Suppose we have trained a Decision Forest using five bootstrapped samples from a data set containing three classes, {Red, Blue, Green}. We then applied the model to a specific test input, $\mathbf{x}$, and observed five estimates of class distributions from the five trees, respectively:

$$\begin{aligned}\{P_1(\text{Red}|\mathbf{x}), P_1(\text{Blue}|\mathbf{x}), P_1(\text{Green}|\mathbf{x})\} &= \{0.1, 0.6, 0.3\} \\ \{P_2(\text{Red}|\mathbf{x}), P_2(\text{Blue}|\mathbf{x}), P_2(\text{Green}|\mathbf{x})\} &= \{0.2, 0.3, 0.5\} \\ \{P_3(\text{Red}|\mathbf{x}), P_3(\text{Blue}|\mathbf{x}), P_3(\text{Green}|\mathbf{x})\} &= \{0.3, 0.5, 0.2\} \\ \{P_4(\text{Red}|\mathbf{x}), P_4(\text{Blue}|\mathbf{x}), P_4(\text{Green}|\mathbf{x})\} &= \{0.2, 0.3, 0.5\} \\ \{P_5(\text{Red}|\mathbf{x}), P_5(\text{Blue}|\mathbf{x}), P_5(\text{Green}|\mathbf{x})\} &= \{0.2, 0.4, 0.4\}\end{aligned}$$

Let us consider the average probability for combining these results into a single class prediction. What will the final class prediction be? *Motivate your answer by short phrases.* (1p)

- e) What is the kind of additional randomness involved in the design of Decision Forests with respect to Bagging? (1p)

A-4 Regression with regularization

(4p)

- a) In regression, regularization is a process of introducing additional term, so-called *shrinkage penalty*. Which ones of the three methods includes the additional term.
- LASSO.
 - Logistic regression.
 - k -NN regression.

Simply indicate your choice. (1p)

- b) For a set of N training samples $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_N, y_N)\}$, each consisting of input vector $\mathbf{x}$ and output y , suppose we estimate the regression coefficients $\mathbf{w}^\top (\in \mathbf{R}^d) = \{w_1, \dots, w_d\}$ in a linear regression model by minimizing

$$\sum_{n=1}^N (y_n - \mathbf{w}^\top \mathbf{x}_n)^2 + \lambda \sum_{i=1}^d w_i^2$$

for a particular value of λ .

Now, let us consider different models trained with different values of λ , starting from a very large value (infinity) and *decreasing* it down to 0. Then, the training error and test error (residual sum of squares, RSS) are believed to change. Indicate which of i. through v. is most likely to be the case. *Justify your answer.* (2p)

- Remain constant.
 - Steadily increase.
 - Steadily decrease.
 - Decrease initially, and then eventually start increasing in a U shape.
 - Increase initially, and then eventually start decreasing in an inverted U shape.
- c) As we *decrease* λ , from infinity, what happens to the variance of the model. Choose the answer from the five alternatives above, and *briefly justify your answer.* (1p)

A-5 Dimensionality Reduction, Subspace Methods

(4p)

We consider to solve a K -class classification problem with the Subspace Method and for that we compute a subspace $\mathcal{L}^{(j)}$ ($j = 1, \dots, K$) using training data for each class, respectively. That is, given a set of feature vectors (as training data) which belong to a specific class C (i.e. with an identical class label), we perform PCA on them and generate an orthonormal basis $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ which spans a p -dimensional subspace, $\mathcal{L}$, as the outcome.

Provide an answer to the following questions.

- a) Given that we compute $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ as eigenvectors of the auto-correlation matrix Q based on the training data, how should we choose the eigenvectors in relation to the corresponding eigenvalues of Q ? (1p)
- b) We have a new input vector $\mathbf{x}$ whose class is unknown, and consider its projection length on $\mathcal{L}^{(j)}$ ($j = 1, \dots, K$). Briefly explain the way to determine the class to which vector $\mathbf{x}$ should belong using the projection lengths. (1p)
- c) Now, describe how the projection length is represented, using a simple formula. (2p)

B Graded problems

Potential inquiries to be addressed to Olov Andersson.

B-1 Probability Theory

(4p)

- a) Assume that you have a joint probability distribution $P(A, B)$. First *write* the formula for Bayes' theorem and then *show how to derive* it. (2p)
- b) Assume you have five random variables A,B,C,D,E. For each of the following expressions 1 to 4 below, state if they are *equivalent* or *not equivalent* to the joint probability distribution $P(A, B, C, D)$:
- 1) $P(A, B, C|D)P(D)$
 - 2) $P(A|B)P(B|C)P(C, D)$
 - 3) $P(C, D|A, B)P(A, B)$
 - 4) $\int_E P(A, B, C, D, E)$ (2p)

B-2 Maximum likelihood estimation for Regression

(4p)

You wish to model a labeled dataset $\mathcal{D} = ((x, y)_i : x, y \in \mathbb{R})$ of N observations with the following model:

$$y = w_0 + w_1 \sin(2x) + \epsilon \quad (1)$$

where $\epsilon \sim \mathcal{N}(0, \sigma^2)$, i.e.,

$$P(\epsilon) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\epsilon^2/2\sigma^2}. \quad (2)$$

Derive a formula for the maximum likelihood estimate of w_0 as a function of w_1 . *Show the steps of your derivations.*

B-3 Maximum likelihood estimation for Classification

(4p)

Consider a binary classification problem with only two data points (for simplicity), where the output class y_i is encoded as 0 or 1:

$$\mathcal{D} = ((x_1, y_1), (x_2, y_2)) = ((1, 0), (2, 1))$$

You decide to fit a simple logistic regression model with parameter θ , defined by:

$$P(y = 1 | x, \theta) = \sigma(\theta x) = \frac{1}{1 + e^{-\theta x}},$$

where $\sigma(\cdot)$ is the logistic sigmoid function.

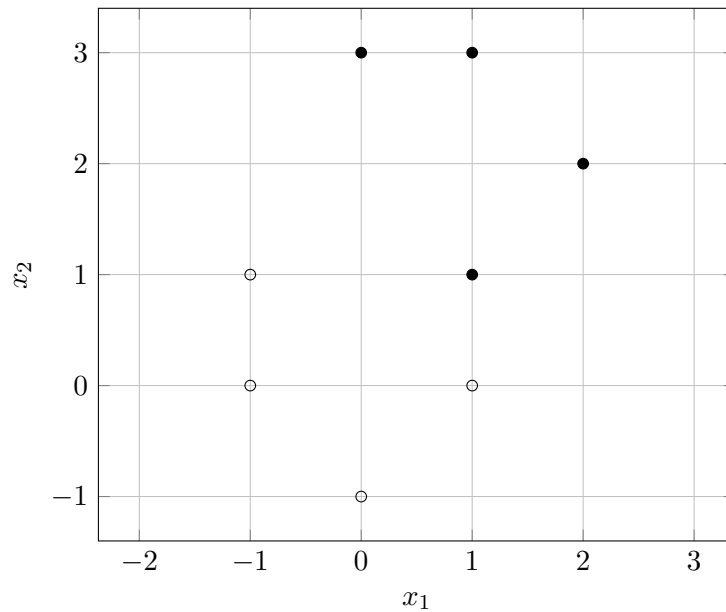
- a) *Write* the likelihood function $L(\theta)$ with the numbers from the given data set. (2p)
- b) Show how one can find the maximum likelihood estimate of θ (you do not need to calculate the value of θ , but concisely explain how to do that). (2p)

C Graded problems

Potential inquiries to be addressed to Jörg Conradt.

C-1 Support Vector Machines

(5p)



The eight datapoints in the figure are used to train a support vector machine. Filled circles are positive samples and unfilled circles are negative samples.

When a linear kernel ($\mathcal{K}(\vec{x}, \vec{y}) = (\vec{x} \cdot \vec{y} + 1)$) was used, and no slack was allowed ($C = \infty$), this resulted in three support vectors, with the corresponding α -values 0.5, 2, and 2.5.

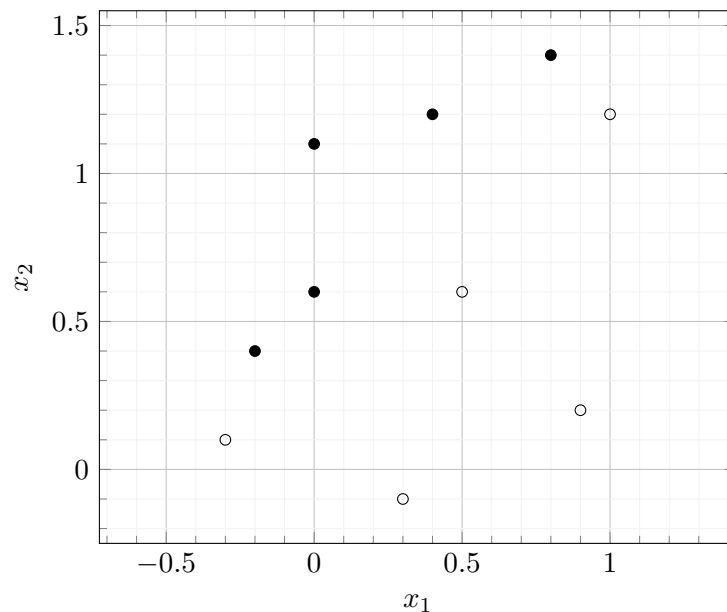
To classify new points, the normal indicator function is used:

$$\text{ind}(\vec{x}) = \sum_k t_k \alpha_k \mathcal{K}(\vec{x}, \vec{s}_k) - b$$

- What will the three support vectors be? You do not need to calculate this mathematically; use your knowledge about support vectors to deduce which three points must be support vectors. (2p)
- Which α -value must be associated with each of the three support vectors? (Give short motivations, ideally keywords) (2p)
- What value for b should be used in the indicator function? (1p)

C-2 Linear classifier, single neuron

(3p)



- a) The figure illustrates a dataset where filled circles are positive samples and unfilled are negative. For a single “artificial neuron”, find a set of values for the weights and the threshold that will separate the positive from the negative samples. Make sure that positive points are *above* the threshold and negative points are *below*.

Give exactly ONE equation that shows the output of the neuron. Ensure that your equation contains all input variables, all weights, and a bias (=threshold) value. (2p)

Note: There is no need to use any learning algorithm here. It is sufficient to find a suitable linear separator in the figure, but you must explicitly state the weight and threshold values and explain how you arrived at these values IN KEYWORDS or short sentences.

- b) If a new negative sample at $(0, 0)$ arrives, and learning is done with the *perceptron learning rule*; how will these values change? (1p)